

Theme A — FINSWARM

(FinNova Capital, digital small-business lending)

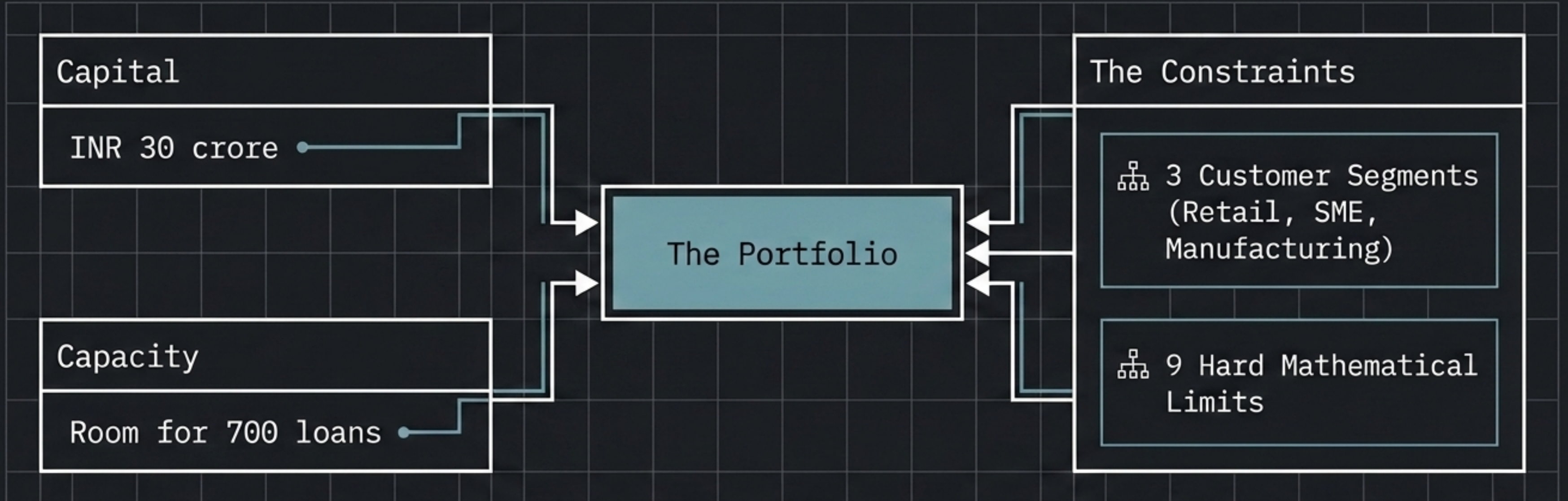
THE AI BOARDROOM

A board that convenes on any business problem.

The Question is Which Portfolio to Actually Write

The problem is not who will default—a credit model answers that.

The problem is portfolio construction with no single owner.



No single department owns the answer. Growth wants volume. Finance wants margin. Risk wants safety. Compliance wants fairness.

The Mandated Board Roster

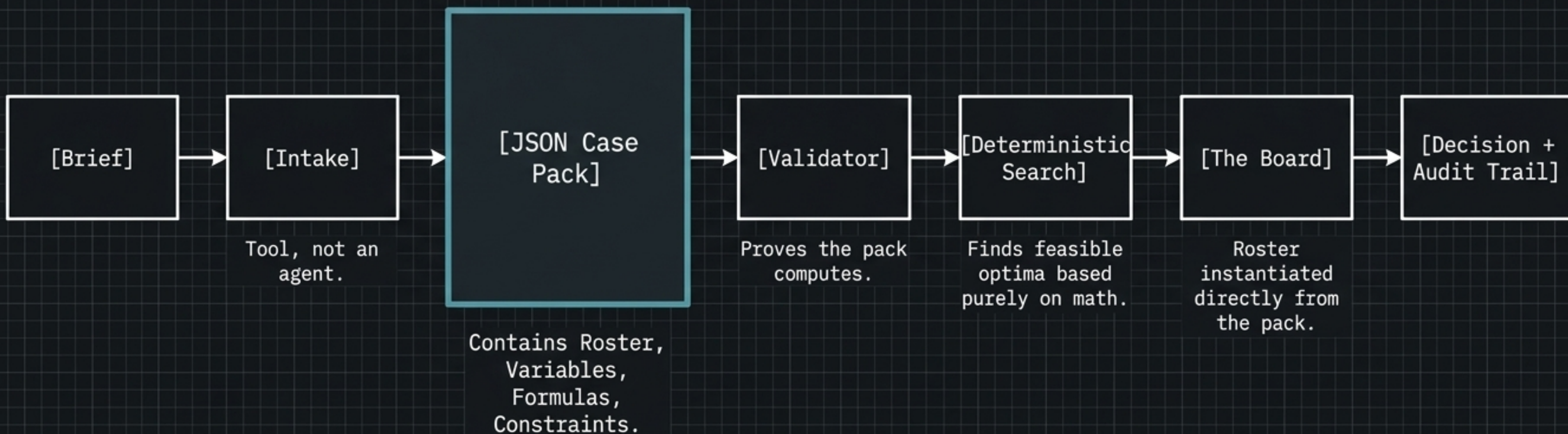
Six executives, each with a defined slice of ownership and mathematical veto authority.

Agent	Owns	Veto Power
Business Research	Market, demand, competitors	Sets the demand ceiling
Finance & Treasury	Cost of funds, liquidity, income	Owns the INR 3 crore liquidity constraint
Credit Risk	Expected losses, portfolio quality	Owns the default cap & 70% concentration cap
Marketing & Sales	Segments, channels, budget	Fights the exclusion of retail shops
Compliance & Customer Protection	Fair treatment, conduct	Owns the 19% interest-rate cap
CEO	Reconcile, decide, sequence	Resolves blocking objections

`Intake module` acts as a deterministic pre-board tool to generate the case pack. It is not a judged agent.

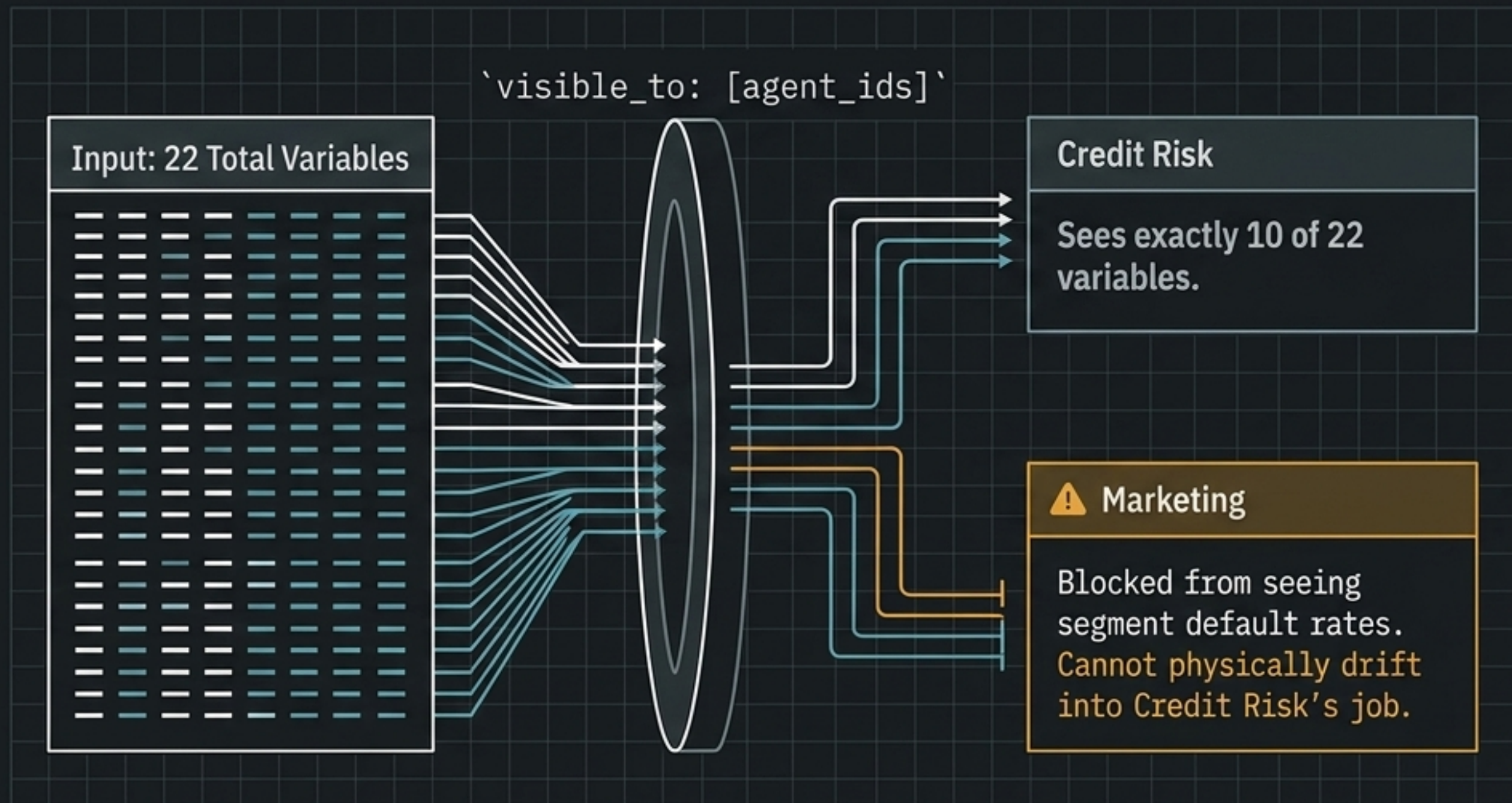
A Domain-Agnostic Engine Pipeline

No Python file knows the industry. The roster, variables, formulas, and constraints live entirely in the JSON pack.



Specialisation is Structural, Not Prompted

Agents cannot reason about variables they cannot see.



This same property does double duty.

On a surprise twist, ``visible_to`` acts as the Impact Map, waking up only the agents whose visible facts moved.

Formulas from the LLM. Arithmetic from a Sandbox.

Every currency figure on screen comes from a whitelisted Python AST evaluator.
Agents never compute.

LLM Output (Text & Signals)

- Emits lever preferences
`n_sme = 306`
- Emits 0-1 signals
`risk_score = 0.72`
- **No arithmetic.**

AST Sandbox (Math & Strict Limits)

- Evaluates expression strings from the JSON pack.
- No attribute access, no imports, no builtins.
- Unknown names rejected before compilation.

The search finds what is feasible. The board decides what is wise.

TC1 Baseline: The Greedy Answer is Illegal

The profit-maximising mix is arithmetically infeasible. Constraints are checked, not weighted.

S_greedy Portfolio

450 Manufacturers + 250 SMEs
(Best margin segments)

Capital Required:
INR 55.5 crore

vs INR 30 crore

REJECTED

Concentration:
~73% Manufacturers

vs 70% cap

REJECTED

The Blockers



Finance: Blocked on Liquidity mandate.



Credit Risk: Blocked on Concentration mandate.

Result: Score 0.
Owning agents raise blocking objections.

TC1 Baseline: The Feasible Optimum Sparks an Argument

The legal portfolio serves zero retail shops, forcing a real, unscripted departmental disagreement.

Marketing & Sales

Position: Files a material objection.

Backs a retail-inclusive alternative on growth grounds to capture the largest demand pool.

Portfolio G1

0 Retail | 306 SME | 96 Manufacturer

Rate: 19% | Net Income: INR 6,537,000

Compliance

Position: Counter-objects.

Higher retail exposure inevitably raises conduct risk. Defends the G1 portfolio.

TC1 Baseline: The CEO Decision

A one-sentence resolution, backed by an eight-part audit trail validated entirely in code.

Decision: G1 Portfolio Chosen.



Customer Segment



Product Terms



Approval Policy



Budget Allocation



Risk Limits



Go-To-Market



Implementation Sequence



Measurable Outcomes (3 KPIs)

Code validation ensures rejected alternatives (and the specific constraint each broke) are explicitly mapped in the final payload.

TC2 Surprise: Selective Re-Run

Credit risk spikes. The impact map wakes up only the agents whose visible facts moved.

The Shock: 4 Invalidated Assumptions

retail_default: 0.05 → 0.08 ⚠

sme_default: 0.035 → 0.05 ⚠

mfg_default: 0.045 → 0.07 ⚠

default_limit: 0.05 → 0.055 ⚠

The Re-Run Map

ACTIVE

Compliance

Credit Risk

Finance

Forced to re-run.

DIMMED

Business Research

Marketing

Did not re-run
(changed variables
structurally invisible).

TC2 Finding: The Pilot is Infeasible As Designed

The deterministic search yields **0 feasible portfolios**. The rules are **jointly unsatisfiable**.

70% Concentration Cap

Forces at least 30% of the book into a secondary segment.

0 Feasible Portfolios

5.5% Default Cap

Secondary segments now default at 7% or worse, mathematically breaching the new portfolio cap.

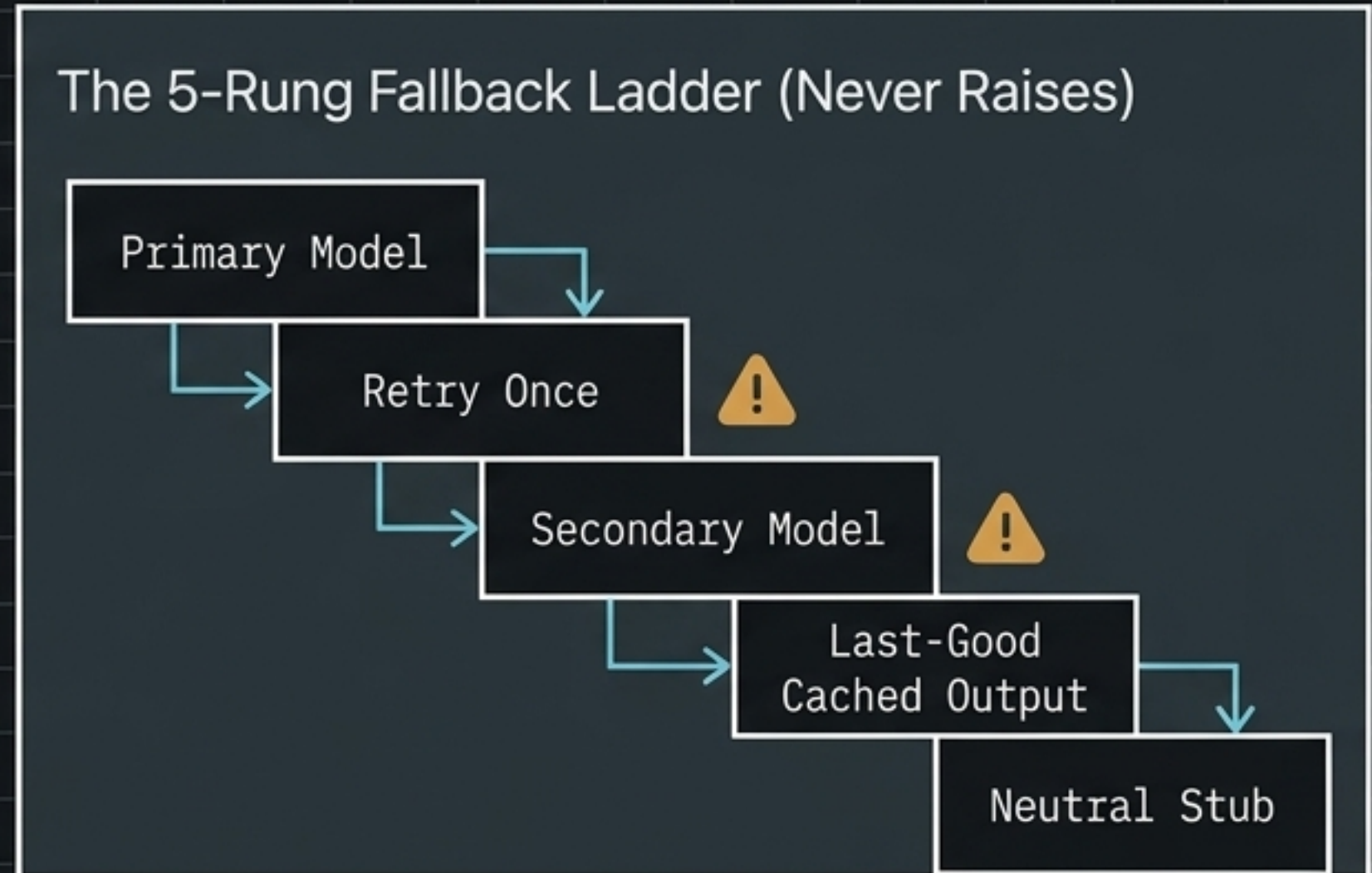
Alternative: An all-SME book clears the 5.5% cap, but outright breaches the 70% concentration rule.

Zero legal portfolios exist. The board forces the CEO to select: Redesign Required.

How It Is Built & What We Claim

Standard libraries, strictly typed data, and a bulletproof degradation path.

The Stack Grid	
Core	Python 3.11, Pydantic, ThreadPoolExecutor
Math	stdlib ast sandbox, stdlib random search + local refinement
LLM & UI	Gemini 3 Flash, Streamlit
Frameworks/Solvers	NONE. Deliberately.



Portfolio optimisation under constraints is textbook. The real contribution is the decision layer on top—constraint ownership, cross-functional objection, and selective re-planning with a full audit trail.